



Stabilising the spike protein

Researchers are investigating methods to stabilise the spike protein, the structures that allow cell infection. <http://digit.in/sep20-61>



CoroFlu vaccine

Bharat Biotech and UW-Madison researchers are one of the teams working towards a vaccine. <http://digit.in/sep20-62>

Special variants of T-cells, called Invariant natural killer T cells (iNKT) offer protection from infections, and at the same time suppress the cytokine storm. The iNKT cells were previously being used to develop treatments for cancer. Researchers from UCLA are also looking into using iNKT cells as a part of treatments for COVID-19, but the cells are incredibly rare, with only 1 cell in a million blood cells being the iNKT variant. Stem cells can be used to produce a large number of iNKT cells, which is now being tested for its efficacy as a treatment.

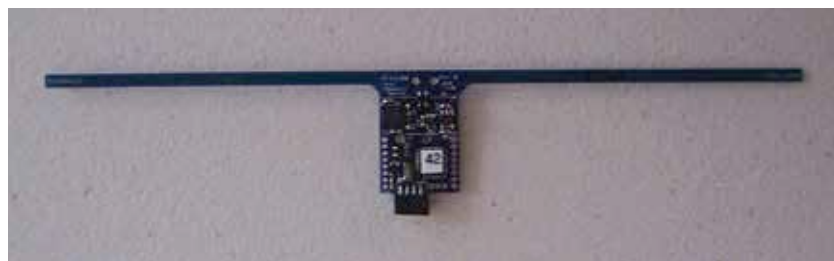
FIGHTING THE VIRUS WITH TECHNOLOGY

One of the most interesting research from MIT may result in a simple smartphone app that test if a person has been infected by the virus using just their voice. This is a more advanced version of how a human can identify if a person has a cold by just listening to their voice. Researchers used voice samples of news anchors and celebrities who were known to have the disease, and trawled YouTube for sound clips. Then, they carefully compared various features from the audio, which served as proxies for the health of the various parts in the body that produce speech. The loudness indicated the health of the respiratory system, the steadiness of pitch indicated the stability of the larynx, and for the other moving parts such as the mouth, tongue and jaw, they extracted speech formants which change in frequency based on the particular shapes and configurations of the parts used to produce the speech. The researchers found that there were in fact some indications that someone was suffering from the onset of the disease, which is similar to a pianist with fingers stuck together trying to play the piano. The researchers are now fine tuning the approach using voice samples from Carnegie Mellon University, a public dataset of audio clips of people who suffered from COVID-19.

Apart from smartphones, another consumer electronics category that can be recruited in the fight against COVID-19 are wearables. Fitness trackers and smartwatches are already equipped with the sensors that can monitor the body for heart rate and skin temperature, two measurements that are known to spike when an infection takes place. Researchers from Stanford are collaborating with FitBit to check if they can analyse the data from fitness trackers in such a way that they can reliably identify signs of early onset of the disease.

X-Ray images of lungs, in an effort to create a diagnostic aid that can identify a patient with the disease, and differentiate between different phases of the infection, and help doctors figure out the best treatment procedure for each individual patient.

With a shortage of respirators, researchers have identified various ways in which the respirators that were supposed to be single use, can be decontaminated and made suitable for reuse. Researchers from UMICH have developed a dedicated chip that can verify if the decontamination



A wireless sensor to check the decontamination of masks

Credit: UMICH

MIT researchers are also using AI to identify how the body is likely to react to different drug cocktails that could potentially be used to treat the disease. Researchers from the University of Michigan are deploying AI to measure the compliance with social distancing using public CCTV cameras, which is something that they can do for any such camera system in the world. The system can be used to track how the pandemic is changing the behavior of people. Another problem that is arising with the progress of the pandemic, is a shortage of drugs right now, and in the future this may occur with vaccine doses. UMICH researchers are also looking into how AI can be deployed to ease these supply issues, and make sure that the drugs and treatments are provided first to those who need it the most. The researchers have identified various ways in which it is possible to make the same drugs cheaper, and bring it to patients quicker, by removing unnecessary steps in the production. Researchers from the University of Chicago are training computers on

process was in fact carried out in a proper way. The sensors are without batteries, are powered wirelessly, and provide an indication when a decontamination process is complete, a bit like a microwave beeping after heating. For the decontamination to work, the respirator must be exposed to just the right amount of temperature and humidity, for the right duration of time. Even if one of these factors are not accurately met, then the respirator may still be unsafe.

MIT and Caltech are both independently using machine learning algorithms to predict how the pandemic is likely to progress once things start opening up again, and identify specific interventions that could halt the spread of the disease. This includes restarting economic activities, mass transit systems, and opening up educational institutions. The studies can also help give healthcare workers advance warning of potential spikes, so that the system can brace for an influx of new patients. These studies will help guide policy decisions. 